

Instructor: Date:

Learner: Regional Branch:

Program: Center:

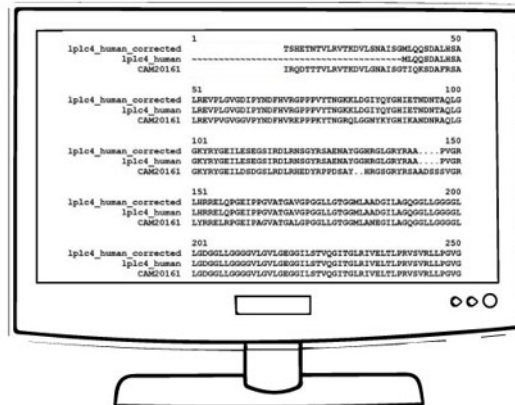
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Let's Start!

A Quick Review on Software Testing and Estimation Techniques



Learning Outcome:

Implement and document the software tests, applying trial and error, based on the designed plan and procedures established by the company.

Applies the different types of software testing to the IT solution.

Can Do Statements:

- I can write a short paragraph describing one software test and a possible estimating technique.
- I can use vocabulary related to software testing and estimating techniques.

Ask your partner:

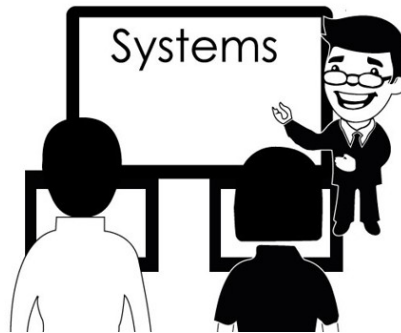
- Why do you think it is important to test software?
- Do you think it is important to consider time when testing a software? Why?





Let's Explore!

If the software is not working, it might need some modifications. Let's test it!



Instructions: Read the text below.

If you haven't been able to get the expected output when trying your software, you might be using the wrong software testing process. As we've said before, software testing, which encompasses different types of testing, is basically the process where we evaluate if a software item is working as expected or not. A software tester, who should carry out the test during the software's development process, could choose among different software testing types depending on what the software requires. For the moment, we are going to talk about the ones we use the most in our company.

To begin with, black box testing, which doesn't consider the evaluation of the internal system design, focuses on requirements and functionality. On the other hand, if a software developer wanted to test the internal logic of an application's code, white box testing could be the most suitable type of test. Furthermore, if the test was carried out by end-users and not only testers, then the software must have gone through a beta testing process which is done before the application is released.

Now, when testing software, we also need to consider another important aspect: Software test estimation, which is one of the most important tasks in test management, is the approximation of how long a task might take to be completed. When using software test estimation, we need to determine how long a testing process may take and how much would it cost. Let's talk about some software test estimation techniques. Work breakdown structure, which is also known as WBS, is the testing process where the project is broken down into smaller pieces to make the evaluation of the software easier. Then, three point estimation, which is normally used after the breaking down structure technique has been carried out, is the method where three types of time estimations are considered (optimistic estimate, pessimistic estimate and realistic estimate). And then, we have Delphi technique, which we use most often, is the method where different experts estimate the time-effort that could take a task to be completed and then, we analyze the feedback from the experts for reaching a final decision on how much time and effort must the test take.



Let's Practice!

Can you recognize information about software testing and estimating techniques?

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. These procedures may allow testers and programmers to carry out different methods in a single software sample.
2. Certain software programs may undergo system testing and beta testing before their release, but might not need to endure the compatibility testing or any other.
3. A good estimation technique could help testers plan the software evaluation.
4. The website content could be seen on the left, and on the right side of the screen, there is a source editor where the HTML source code appears.
5. Most of free online editors might offer the same features as paid versions, except for variety of color scheme or uploading options.

Information related to software testing and estimating techniques	Information that has nothing to do with software testing and estimating techniques.



Let's Practice!

Choosing the best software testing for your project is highly important.

Instructions: Organize the mystery sentence. Read the sentences aloud. Check your pronunciation with your instructor.

1. different types of testing, / Software testing, / software performance. / which encompasses evaluates
2. Software testers, / choose among the types / development process, / who assess a software / during the / of testing.
3. the internal system design, / which doesn't evaluate / Black box testing, / requirements and functionality. / focuses on
4. into smaller pieces. / Work breakdown structure, / is the process where / also known as WBS, / the project is broken down
5. which we often use, / experts estimate testing time / is the method where / and effort. / Delphi technique,



Extension Activity

How would you design a software project?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Think about a software project that would have a positive impact in your community.
- List the resources, human skills, time and cost that you might need.
- Write a short project presentation in which you explain the stages for the development of your project.
- Present your project to your classmates.

Self-Assessment

Yes

No

- I think about a software project.
- I make a list of the resources, human skills, time and cost that I might need.
- I write a project presentation.
- I present my project to my classmates.

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Target Words

- encompass** (verb): to include an element or elements in a group of things that share certain characteristics.
e.g. Software testing, which encompasses different types of testing, is basically the process where we evaluate if a software item is working as expected or not.
- break down** (verb): to divide something into smaller parts, to subdivide.
e.g. Work breakdown structure, which is also known as WBS, is the testing process where the project is broken down into smaller pieces to make the evaluation of the software easier.
- functionality** (noun): it refers to the functions or uses of something.
e.g. Black box testing, which doesn't consider the evaluation of the internal system design, focuses on requirements and functionality.

- suitable** (adjective): being appropriate for a given purpose
e.g. If a software developer wanted to test the internal logic of an application's code, white box testing could be the most suitable type of test.
- feedback** (noun): the return of information that was analyzed by someone or something in order to improve a process
e.g. Delphi technique, which we use most often, is the method where different experts estimate the time-effort that could take a task to be completed and then, we analyze the feedback from the experts for reaching a final decision on how much time and effort must the test take.



Learning Points

Modal Verbs for Speculating

There are **modal verbs** used to make assumptions about something.

Modal verbs can indicate speculation when the speaker wants to express that something is likely to be happening or happen in a certain way or for a given reason.

Could, may, must and **might** are modal verbs used to speculate. They usually have different degrees of possibility. **Must** expresses that something is very likely to take place in a certain manner. **Could** expresses a less likely probability; and **may** and **might** are used when the speaker is very uncertain about the likelihood of a situation.

Examples:

*Any project **may** take more time than it was estimated.*

*Estimation techniques **could** help you to organize and re organize the work when it is needed.*

*These software tests **must** help us identify new errors.*

Examples

- *If you haven't been able to get the expected output when trying your software, you might be using the wrong software testing process.*
- *If the test was carried out by end-users and not only testers, then the software must have gone through a beta testing process which is done before the application is released.*
- *If a software developer wanted to test the internal logic of an application's code, white box testing could be the most suitable type of test.*

Subject and Object Relative Clauses: Non-Defining

Relative clauses are used to introduce extra information that might be relevant or not for the meaning of the sentence.

Non-defining relative clauses are different from defining relative clauses as they introduce additional information that **is not crucial** for the understanding of the whole sentence. The pronouns used are **which** if we refer to an **object** and **who** if we refer to a **person**.

We can write the relative clause between commas if the relative pronoun comes after the subject.

Example:

*Brand computers, **which** are expensive, make the best for office work.*





Learning Points

Or after a comma if the relative pronoun comes after an object:

Example:

*I bought a brand new computer, **which** was very expensive.*

We can often also leave the relative pronoun out as long as it is the object of the verb.

Example:

*This is the software **I used last season.** 4*

Examples

- *Software testing, which encompasses different types of testing, is basically the process where we evaluate if a software item is working as expected or not.*
- *A software tester, who should carry out the test during the software's development process, could choose among different software testing types depending on what the software requires.*
- *Three point estimation, which is normally used after the breaking down structure technique has been carried out, is the method where three types of time estimations are considered (optimistic estimate, pessimistic estimate and realistic estimate).*